

Arrhenius law for the eigenvalues of the Smoluchovski operator

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In this tutorial, we want to see how the Arrhenius law emerges from the overdamped Langevin equation. The Arrhenius law states that it takes a time $\tau \propto \exp(-U/T)$ for an overdamped Langevin process to cross an energy barrier E . First, we have to define rigorously the time τ . Second, we aim at computing this time, maybe numerically, in a given potential double-well $V(x)$ for a one-dimensional process. Third, we may study in more details the properties of the solutions.

1 Definition of the barrier crossing time

1. Give an example of a confining double-well potential $V(x)$.
2. Write the Smoluchovski equation for a one-dimensional overdamped Langevin process in the potential $V(x)$ as

$$\partial_t p(x, t) = -\mathcal{L}_S p(x, t), \quad (1)$$

where $\mathcal{L}_S$ is the Smoluchovski operator, which acts only on x . What are the properties of the spectrum of $\mathcal{L}_S$? What is the first eigenvalue in ascending order, ω_0 , and what is the corresponding eigenmode?

3. Explain why, in a double-well potential, we may define the crossing time as $\tau = \omega_1^{-1}$, where ω_1 is the second eigenvalue of $\mathcal{L}_S$. Describe qualitatively the spectrum that we expect to find.
4. Explain qualitatively why the potential needs to be confining for the crossing time to be well defined.

2 Calculation of the crossing time

We now aim at computing the crossing time, which should be done numerically in general. To do so, we need to discretize numerically the Smoluchovski operator.

5. Explain qualitatively how to discretize such operator in order to be able to compute its eigenvalues. What approximations need to be made, and how to check that the result of the calculation is correct?

We add another requirement to the discretization: its first eigenvalue and the corresponding eigenvector should match their analytical values at any level of discretization.

6. Explain why this requirement is met if the discretized form of the Smoluchovski equation takes the form of a master equation with detailed balance.
7. Determine all the parameters of the master equation for a given level of discretization (choice of discretization step a and discretization bounds $x_{\min}$ and $x_{\max}$) that correspond to the original overdamped Langevin process.
8. Write a numerical code that takes a potential $V(x)$ and discretization parameters as inputs and returns the spectrum and eigenmodes.

3 Analysis of the results

9. Analyze the spectrum. Show that the crossing time follows an Arrhenius behavior.
10. Plot the first eigenmodes. Under what conditions can the system be mapped on a two-dimensional system? How can we perform such mapping? You can use analogies with quantum mechanics.
11. Explore what happens in a potential with more than two wells.